

3a

A transverse wave is a wave in which the displacement of the particles in the wave are at right angles to the direction of transfer of energy of the wave.

A longitudinal wave is a wave in which the displacement of the particles in the wave are along the direction of transfer of energy of the wave.

3bi

$$\Delta t = 0.78 - 1.5(0.50) \\ = 0.03 \text{ ms}$$

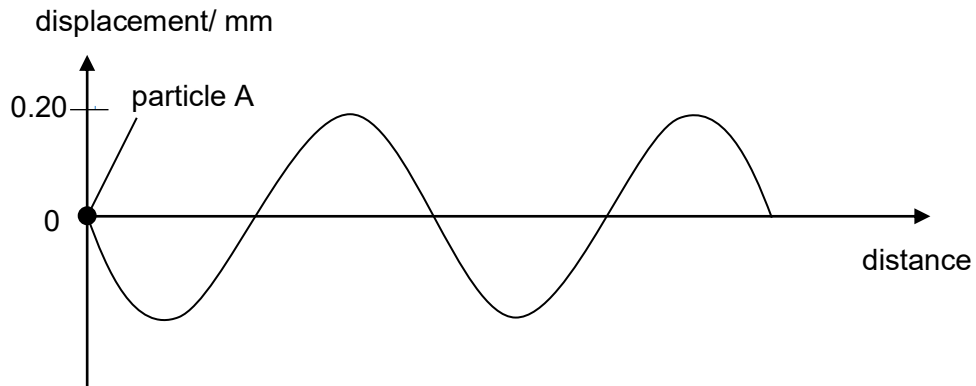
$$\text{OR } \Delta t = 0.50 - (0.78 - 1.5(0.50)) \\ = 0.47 \text{ ms}$$

$$\begin{aligned} \text{Phase difference} &= \Delta t / T \times 360^\circ \\ &= 0.03 / 0.50 \times 360^\circ \\ &= 21.6 \\ &= 22^\circ \end{aligned}$$

OR

$$\begin{aligned} \text{Phase difference} &= 0.47 / 0.50 \times 360^\circ \\ &= 338^\circ \end{aligned}$$

3bii



Correct shape of graph
Correct labelling of amplitude

4a

Using